Layer 3 Switch Inter VLAN Routing

Task 1: On a Cisco Layer 3 switch, create three VLANs (e.g., VLAN 10 for 'Sales', VLAN 20 for 'Support', VLAN 30 for 'Marketing') and assign at least one interface to each VLAN.

Create three VLANs (VLAN 10 Sales, VLAN 20 Support, VLAN 30 Marketing) and assign at least one interface to each.

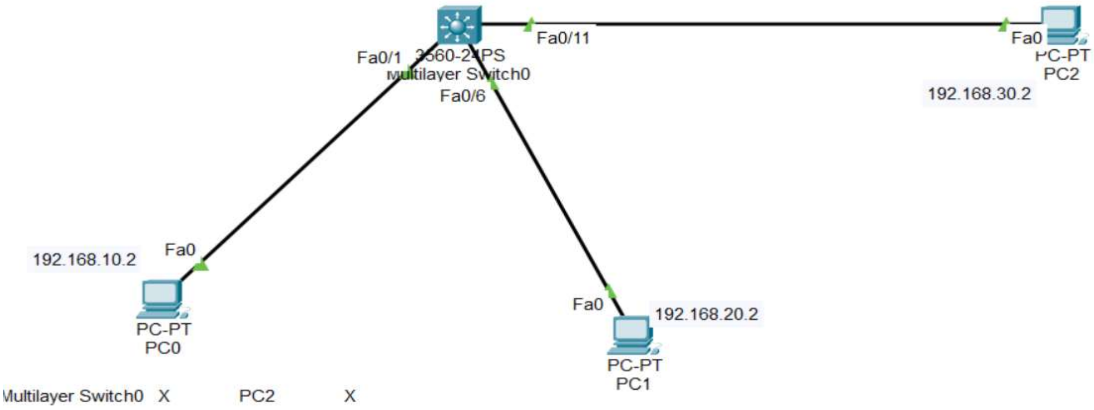
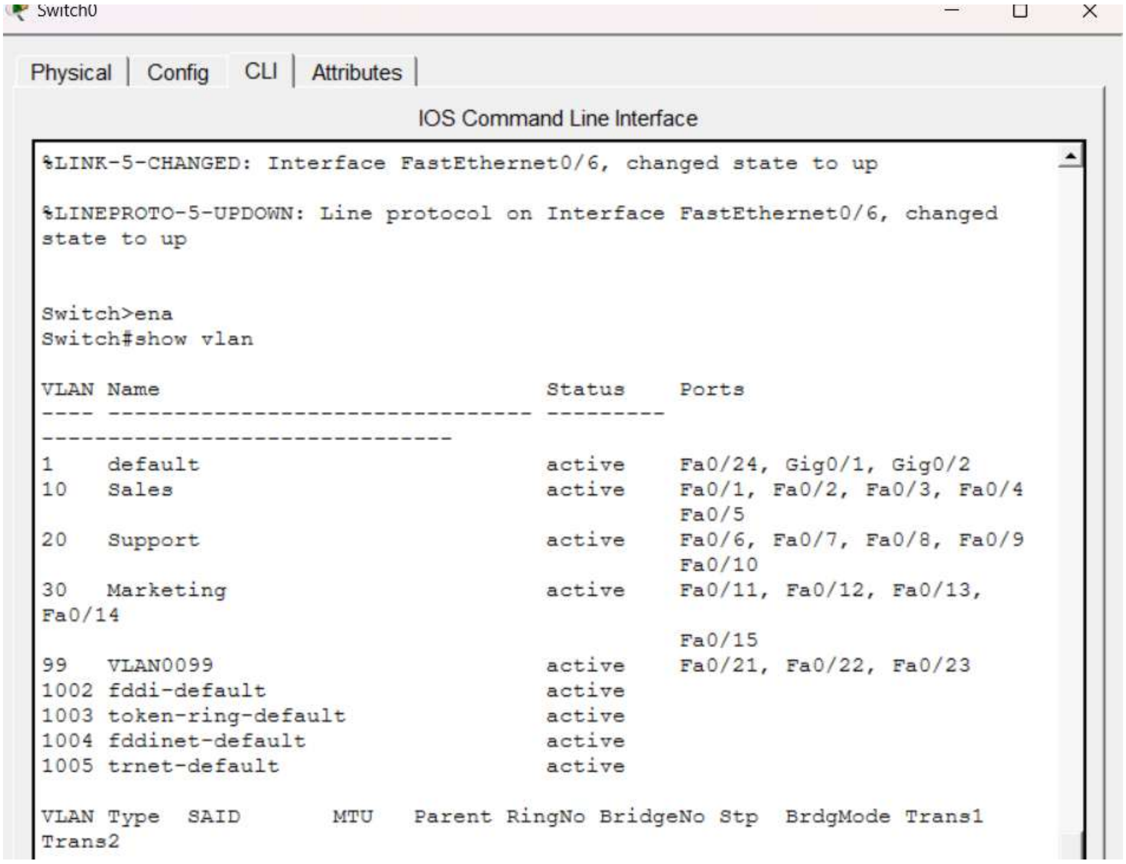


Figure 1: Network topology — PC0, PC1, PC2 connected to Multilayer Switch0

Verification with show vlan confirmed all three VLANs active with the correct ports assigned.

Three PCs were connected to a Cisco 3560 multilayer switch (Multilayer Switch0), one per VLAN:

- PC0 — 192.168.10.2 — VLAN 10 (Sales) — Fa0/1
- PC1 — 192.168.20.2 — VLAN 20 (Support) — Fa0/6
- PC2 — 192.168.30.2 — VLAN 30 (Marketing) — Fa0/12




```
%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed
state to up


Switch>ena
Switch#show vlan

VLAN Name                             Status    Ports
---- -------------------------------- ---------
-------------------------------
1    default                          active    Fa0/24, Gig0/1, Gig0/2
10   Sales                            active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5
20   Support                          active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10
30   Marketing                        active    Fa0/11, Fa0/12, Fa0/13,
Fa0/14
                                                Fa0/15
99   VLAN0099                         active    Fa0/21, Fa0/22, Fa0/23
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1
Trans2
```

Figure 2: show vlan output confirming VLAN 10, 20, 30 active with assigned ports

Task 2: Configure Switch Virtual Interfaces (SVIs) for each VLAN you created by assigning them unique IP addresses (e.g., 192.168.10.1/24 for VLAN 10, etc.) and ensure each SVI is in 'up/up' state. Hint: Use the 'interface vlan' command to create SVIs and 'no shutdown' to enable them.

Configure SVIs for each VLAN with unique IP addresses and confirm up/up state.

```
Switch(config)# interface vlan 10
Switch(config-if)# ip address 192.168.10.1 255.255.255.0
Switch(config-if)# no shutdown

Switch(config)# interface vlan 20
Switch(config-if)# ip address 192.168.20.1 255.255.255.0
Switch(config-if)# no shutdown

Switch(config)# interface vlan 30
Switch(config-if)# ip address 192.168.30.1 255.255.255.0
Switch(config-if)# no shutdown
```

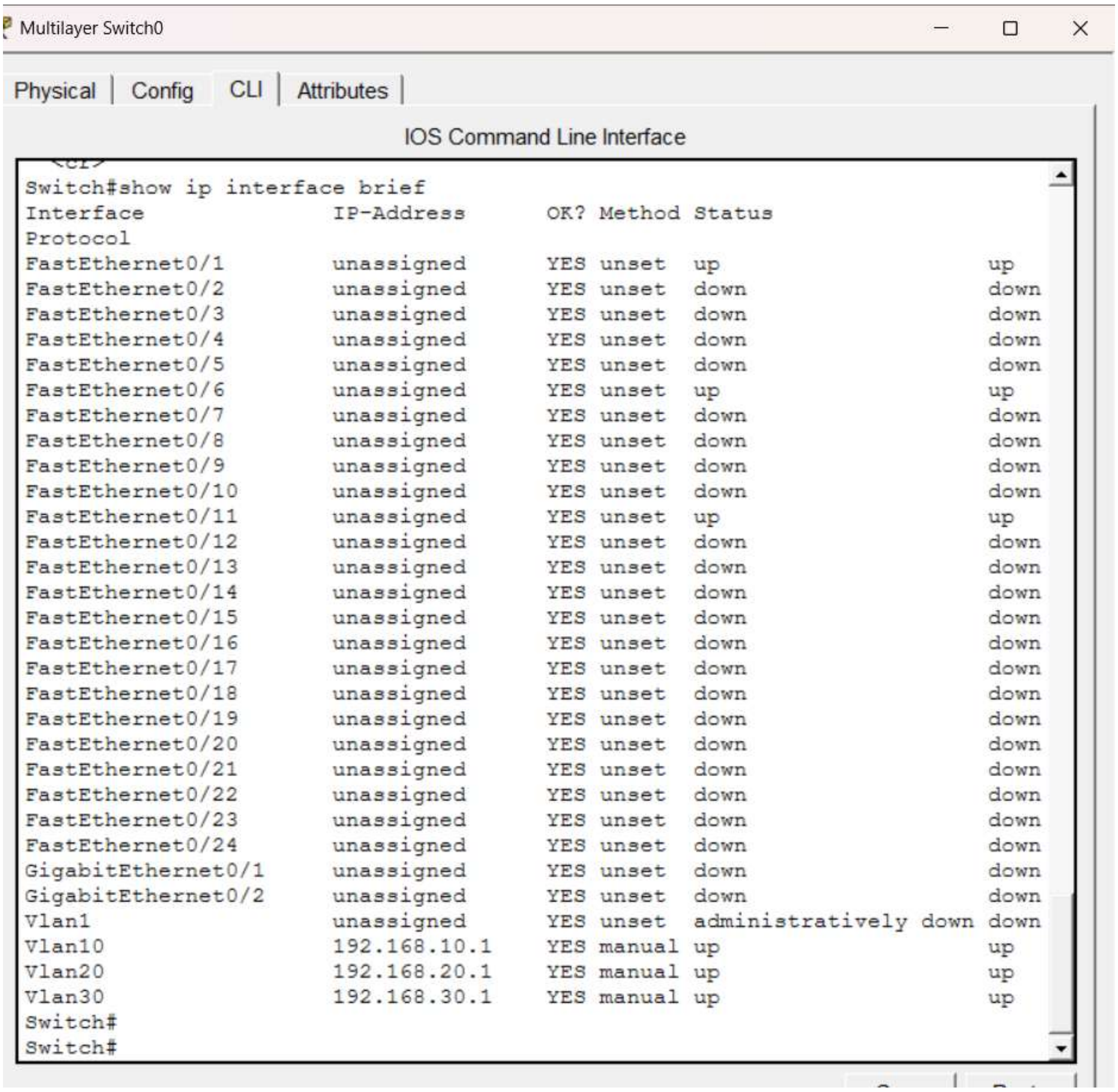


Figure 3: show ip interface brief — Vlan10, Vlan20, Vlan30 all up/up with assigned IPs

Task 3: Enable IP routing on the Layer 3 switch so that devices in different VLANs can communicate with each other without using an external router. Hint: Use the 'ip routing' global configuration command

```
Switch(config)# ip routing
```

This single global command activates Layer 3 forwarding between the configured SVIs, turning the switch into an inter-VLAN router.

Task 4: Verify inter-VLAN routing by connecting two PCs (one in VLAN 10 and one in VLAN 20) and pinging between them. Capture and submit the ping results to show successful communication.

PC0 (VLAN 10, 192.168.10.2) was configured with gateway 192.168.10.1, and PC1 (VLAN 20, 192.168.20.2) with gateway 192.168.20.1. A ping from PC0 to PC1 was then run:

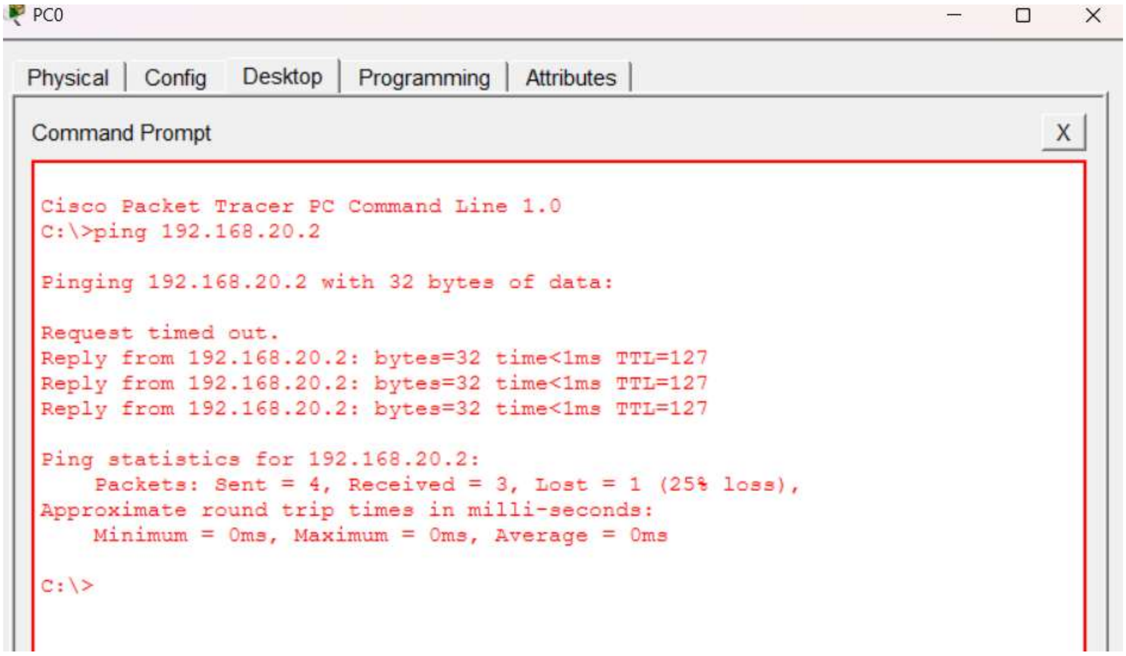


Figure 4: PC0 pinging PC1 (192.168.20.2) — 3/4 replies received, 1 timeout

The first packet timed out due to ARP resolution delay — PC0 had to first send an ARP request to learn PC1's MAC address before the ICMP echo could complete, consuming the timeout window on packet 1. Packets 2–4 succeeded immediately once the ARP entry was cached, confirming inter-VLAN routing was functioning correctly.

Task 5: Refactor your VLAN and SVI configuration so that VLAN 30 uses a different subnet mask (e.g., /25 instead of /24). Explain what changes you made and how this affects communication with the other VLANs.

```
Switch(config)# interface vlan 30
Switch(config-if)# no ip address 192.168.30.1 255.255.255.248
Switch(config-if)# ip address 192.168.30.1 255.255.255.252
Switch(config-if)# no shutdown
Switch(config-if)# do write
```

PC2's static IP configuration was updated to match the new /30 mask (255.255.255.252), keeping its address at 192.168.30.2 and gateway at 192.168.30.1.

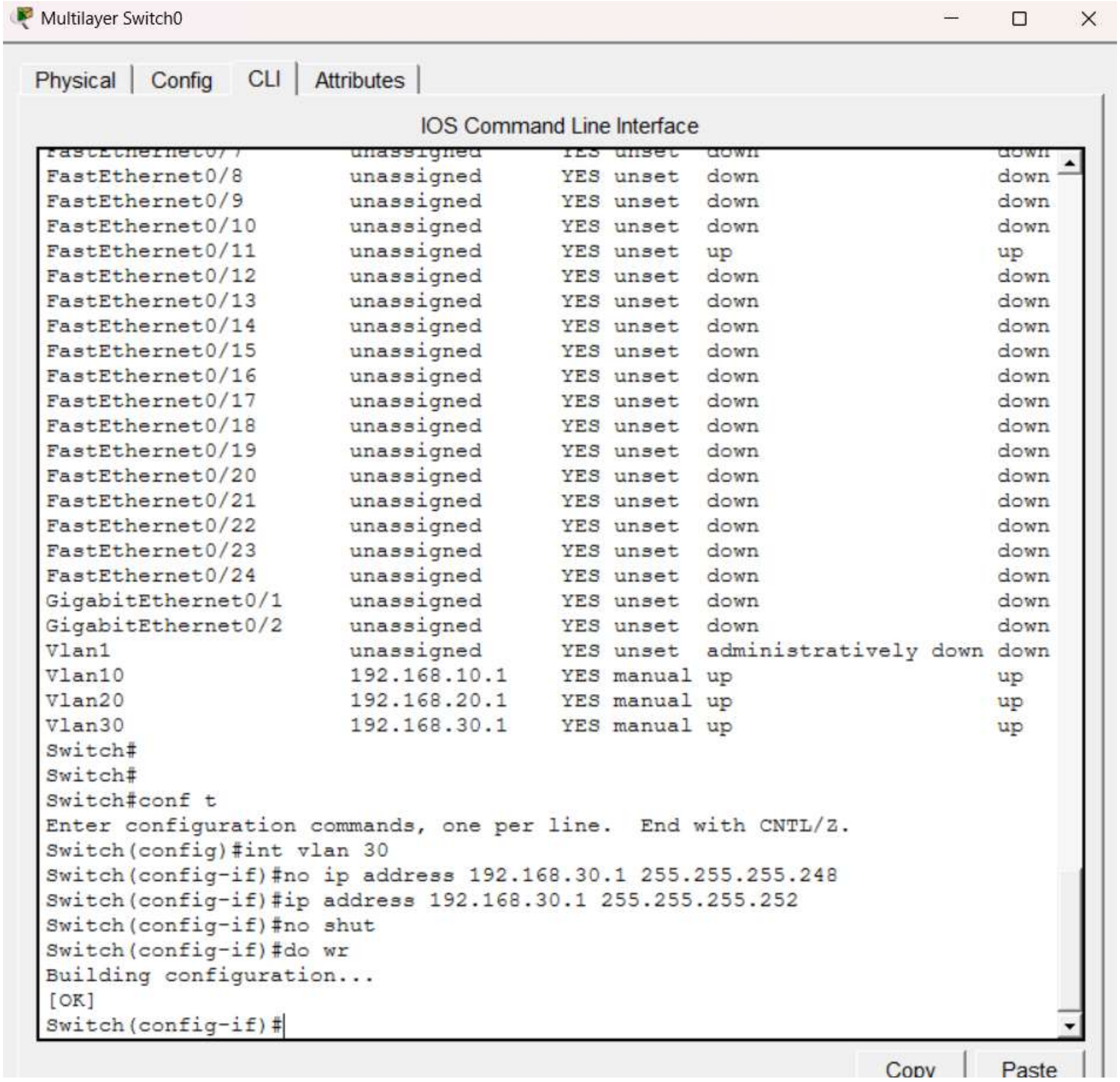


Figure 5: VLAN 30 SVI reconfigured to /30 (255.255.255.252) and saved

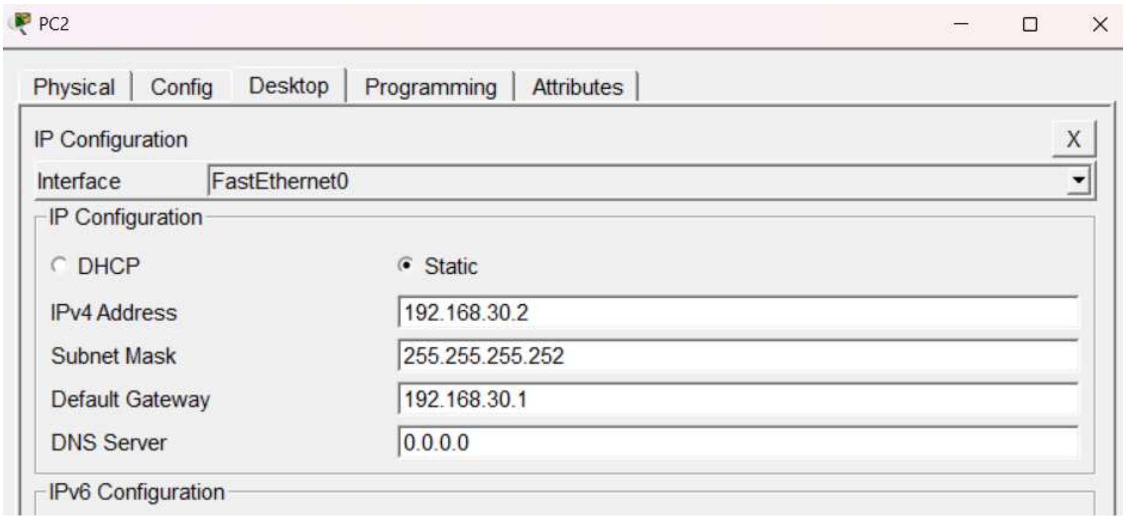


Figure 6: PC2 IP configuration updated to 255.255.255.252 subnet mask

A ping from PC2 to PC0 (192.168.10.2) across the new /30 subnet confirmed full connectivity, with 4 of 4 packets received and 0% loss:

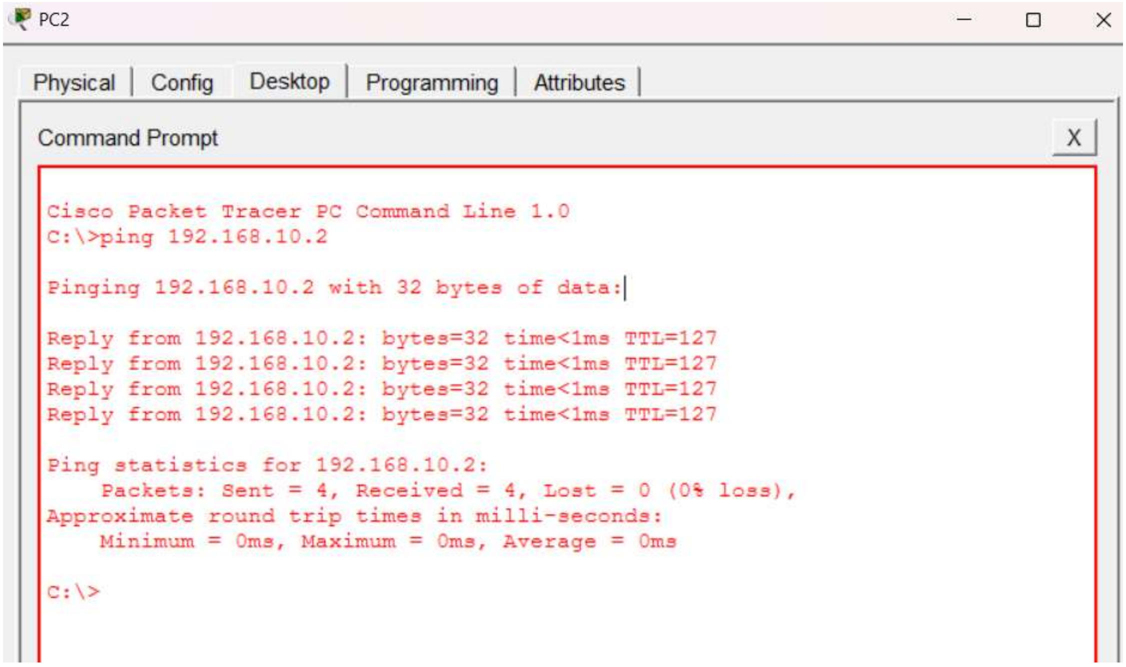


Figure 7: PC2 successfully pinging PC0 across VLANs after the /30 refactor — 4/4 received, 0% loss

Impact of the mask change

- A /30 mask provides exactly 2 usable host addresses, sufficient for the gateway (192.168.30.1) and PC2 (192.168.30.2) on this small, isolated VLAN — but it leaves no room to add further devices without re-addressing.
- VLAN 10 and VLAN 20 were unaffected by the change, since each SVI operates as an independent Layer 3 interface — the switch routes based on each SVI's own subnet, not a shared configuration.
- The only requirement for the change to work was keeping PC2's subnet mask in sync with the SVI's mask; a mismatch between the two would cause PC2 to miscalculate which addresses are local versus which require the gateway, breaking connectivity.